Problem # 1:

Plot and label the C-V curve of a PMOS capacitor with $P^+$ gate, substrate doping of $N_D = 5\times10^{17}$ $cm^{-3}$ and $SiO_2$ = 2 nm.

$$C_{ox} = \frac{\epsilon_{ox}}{d} = \frac{3.9 \times 8.85 \times 10^{-14}\ F/cm}{2\times10^{-7}\ cm} = 1.726\times10^{-6}\ F/cm$$

$$|\phi_F| = \frac{k_B T}{q} \ln\left(\frac{N_D}{n_i}\right) \approx 0.45\ V$$

note: $V_{FB} > 0 \longleftrightarrow$ must "drag down" Fermi level on gate to reach flat band ($\xi = 0$).

$$V_{FB} = \frac{1}{q}\underbrace{(q\chi_{si} + E_g)}_{\phi_{gate}} - \frac{1}{q}\underbrace{(q\chi_{si} + E_g/2 - q|\phi_F|)}_{\phi_{subs}}$$

$$V_{FB} = \frac{1}{q}\left(E_g/2 + |\phi_F|\right) \approx 0.55\ V + 0.45\ V = 1V$$

$$V_T = V_{FB} - 2|\phi_F| - \frac{1}{C_{ox}}\sqrt{2qN_D\epsilon_{si}|2\phi_F|} = 1V - 0.9V - \frac{\sqrt{2\times1.6\times10^{-19}\ C\times5\times10^{17}\ cm^{-3}\ \epsilon_0\epsilon_{si}}}{1.726\times10^{-6}\ F/cm}$$

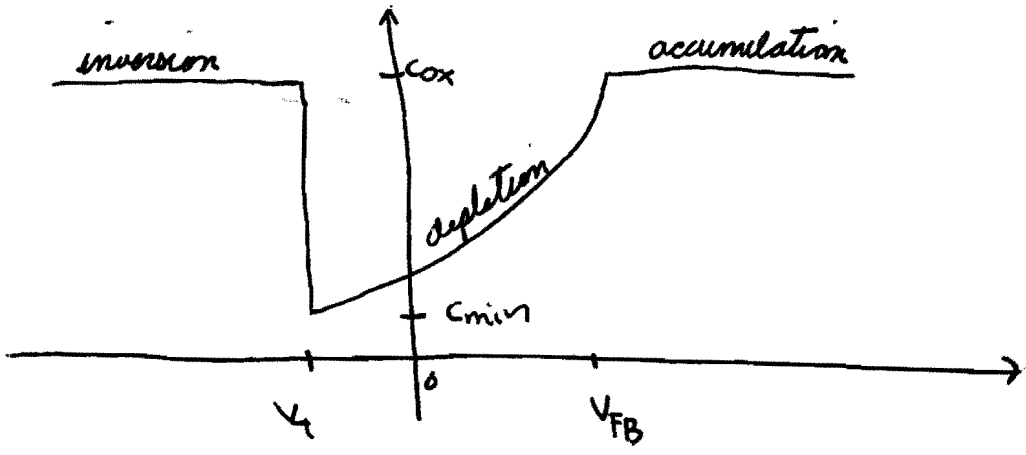
$$V_T = 1V - 0.9V - 0.226\ V = -0.126\ V$$

$C_{ox}$ and $C_d$ in series

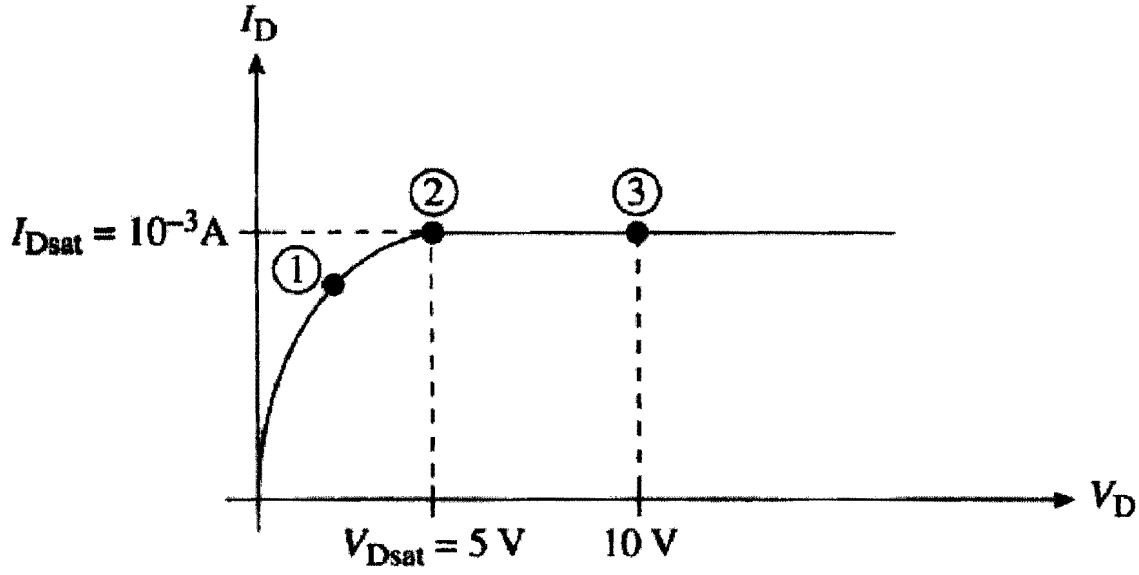
$$C_{max} = C_{ox} = 1.726\times10^{-6}\ F/cm$$

$$C_{min} = \left(\frac{1}{C_{ox}} + \frac{W_{max}}{\epsilon_{si}}\right)^{-1}$$

$$C_{min} = \left(\frac{1}{C_{ox}} + \sqrt{\frac{2|2\phi_F|}{qN_D\epsilon_{si}}}\right)^{-1} = 1.923\times10^{-7}\ F/cm$$



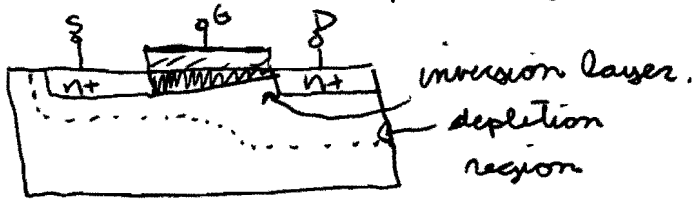
Problem # 2:




An $I_D$ - $V_D$ characteristic derived from an ideal MOSFET is shown above. Note that $I_{Dsat} = 10^{-3}$ A and $V_{Dsat} = 5$ V for the given characteristic.

a. Sketch the inversion layer and depletion region inside the MOSFET corresponding to point 1 on the graph. Show and label all parts of the transistor.

Since $V_D > 0$ → N-channel. @ point #1 the MOSFET is biased below saturation and the channel is not pinched off.




b. Given a turn-on voltage of $V_T = 1$ V, what is the gate voltage one must apply to the MOSFET gate to obtain the pictured characteristic?

We use square law theory: $V_{D_sat} = V_g - V_T$

$$V_g = V_{Dsat} + V_T = 6\,V$$

c. if $x_0 = 0.1$ μm, what is the inversion layer charge / cm$^2$ at the drain end of the channel when the MOSFET is biased at point 2? Point # 2 corresponds to pinch off point. If this is true then

$$Q_N(L) = 0\ C/cm^2$$

d. Suppose the gate voltage is readjusted so that $V_G - V_T = 3$ V. For the new condition, determine $I_D$ if $V_D = 4$ V. $V_D > V_{D_sat} = V_g - V_T \rightarrow$ MOSFET is in saturation

$I_{D_sat} \propto (V_g - V_T)^2$

$I_D = I_{d_sat}' (V_{g1} - V_T = 3V)$

$I_{d_sat2} = 10^{-3} A \ (V_{g2} - V_T = 5V)$

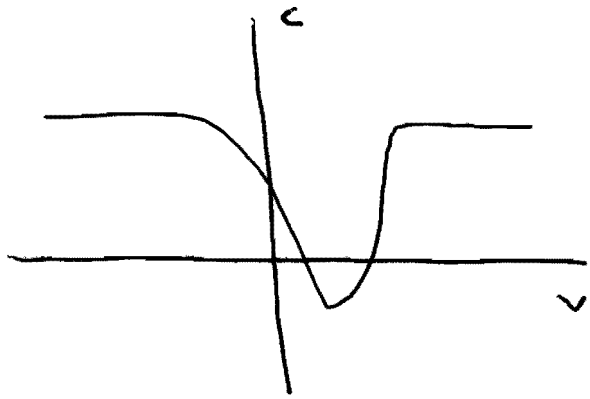
$$\frac{I_{d_sat}'}{I_{d_sat}^2} = \left(\frac{V_{g1} - V_T}{V_{g2} - V_T}\right)^2 = (10^{-3})\left(\frac{3}{5}\right)^2$$

$$= 3.6 \times 10^{-4} A$$

e. If $V_D = 0$ (i.e. the drain is shorted to the source and back), sketch the general shape of $C_G$ (gate capacitance) versus $V_G$ to be expected from the MOSFET.

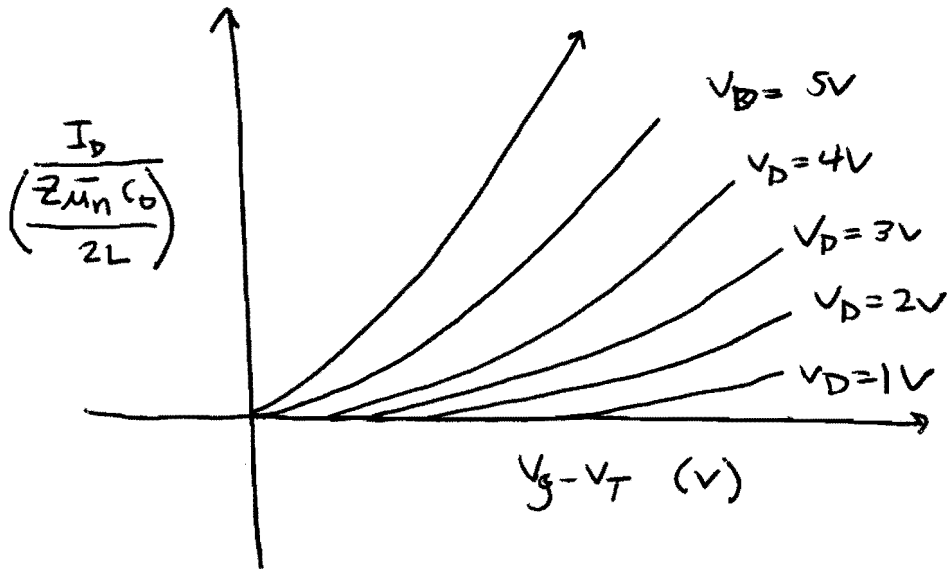
n-channel

low frequency.



Problem # 4:

The most widely encountered MOSFET characteristics are a plot of $I_D$ versus $V_D$ with $V_G$ or $V_G$-$V_T$ held constant at select values. An alternative plot of $I_D$ versus $V_G$ or $V_G$-$V_T$ with $V_D$ held constant at select values is sometimes useful. Sketch the shape of the $I_D$ versus $V_G$-$V_T$ characteristics to be expected from an ideal n-channel MOSFET. Specifically show the characteristics corresponding to $V_D$ = 1, 2, 3, and 4 V. Explain how you arrived at the sketch.




Things to note:

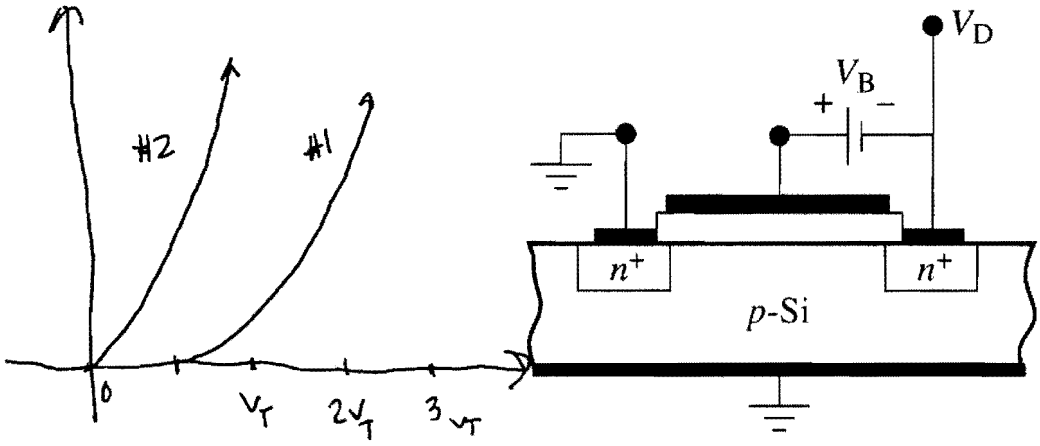
i) increase $V_g - V_T$ from zero w/ $V_D$ = constant initially $V_D > V_g - V_T$ so the device is saturated $I_D = I_{D\text{-sat}} = \frac{Z\bar{\mu}_n C_o}{2L}(V_g - V_T)^2$

ii) when $V_g - V_T = V_D$ → linear region

$$I_D = \frac{Z\bar{\mu}_n C_o}{L}\left[(V_g - V_T)V_D - V_D^2/2\right]$$

iii) increase $V_D$ stay on squared part of curve longer $V_g - V_T > V_D$ linear region w/ increasing slope w/ increasing $V_D$.

Problem # 5:




Suppose a battery $V_B > 0$ is connected between the gate and drain of an ideal n-channel MOSFET as pictured. Sketch the $I_D$ versus $V_D$ ($V_D > 0$) if $V_B = V_T/2$ and $2\,V_T$.

condition #1 : $V_g = V_D + V_B$ or $\boxed{V_D = V_g - V_B}$

in square law theory : $V_{D,sat} = V_g - V_T$. if $V_B = V_T/2 \rightarrow V_D = V_g - V_T/2 > V_{Dsat}$

Therefore the MOSFET is always in saturation. We note $I_D = 0$ if $V_g < V_T$

or if $V_D < V_T/2$ :

$$\boxed{I_D = \frac{Z \bar{\mu}_n C_o}{2L}(V_g - V_T)^2 = \frac{Z \bar{\mu}_n C_o}{2L}(V_D - V_T/2)^2 \qquad V_D > V_T/2}$$

$$\boxed{I_D = 0 \qquad V_D < V_T/2}$$

condition #2 : $V_B = 2V_T \rightarrow V_D = V_G - 2V_T < V_g - V_T = V_{Dsat}$

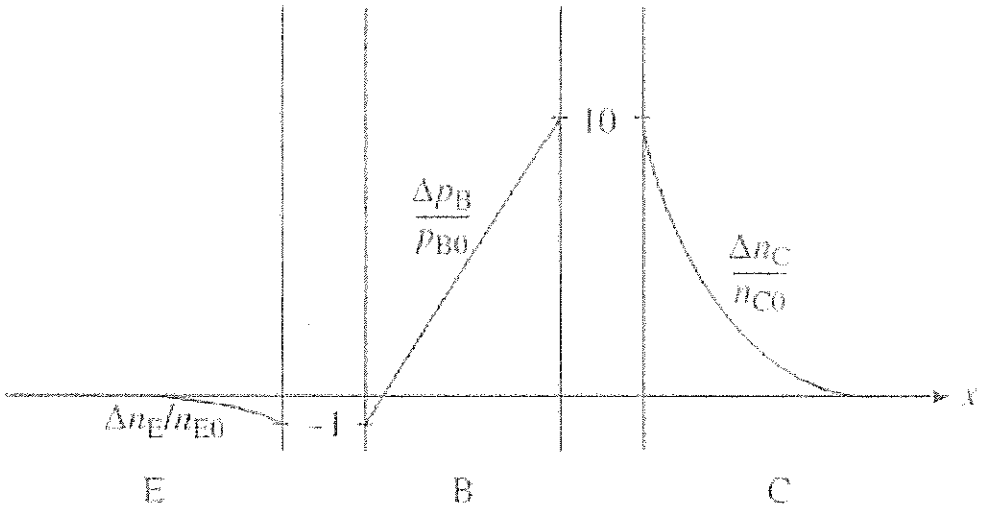
MOSFET is in the linear regime and turns on for $V_g > V_T$ or $V_D > -V_T$

so it is on for all $V_D \geq 0$

$$I_D = \frac{Z \bar{\mu}_n C_o}{L}\left[(V_g - V_T)V_D - V_D^2/2\right] = \frac{Z \bar{\mu}_n C_o}{L}\left[(V_{Dt} V_T)V_D - V_D^2/2\right]$$

$$= \frac{Z \bar{\mu}_n C_o}{L}\left(V_D^2/2 + V_T V_D\right) = \boxed{\frac{Z \bar{\mu}_n C_o}{2L}\left[(V_D + V_T)^2 - V_T^2\right] \quad V_D > 0}$$

Problem # 5:




The $\Delta n_E/n_{E0}$, $\Delta p_B/p_{B0}$ and $\Delta n_C/n_{C0}$ distributions in the quasineutral regions of a *pnp* BJT are shown above.

a. What is the polarity of $V_{EB}$?

$V_{EB} < 0$ as both $\Delta n_E/n_{E0}$ & $\Delta p_B/p_{B0}$ are less than zero at the edge.

b. What is the polarity of $V_{CB}$?

$V_{CB} > 0$ as both $\Delta p_B/p_{B0}$ & $\Delta n_C/n_{C0}$ are greater than zero.

c. What is the magnitude of $V_{CB}$?

$$\Delta p_B(w) = p_{B0}\left(e^{\frac{qV}{kT}} - 1\right)$$

$$\frac{\Delta p_B(w)}{p_{B0}} = e^{\frac{qV_{CB}}{kT}} - 1 = 10$$

$$V_{CB} = \left(\frac{kT}{q}\right)\ln(11) = 0.062\,V$$

Problem # 6:

Complete the table below by indicating whether the noted change in a BJT device parameter increases, decreases or has no effect on the listed performance parameters.

| Change | Effect on $\gamma$ | Effect on $\alpha$ | Effect on $\beta$ |
|---|---|---|---|
| Increase $W_B$ | ↓ | ↓ | ↓ |
| Increase $\tau_B$ | N.C. | ↑ | ↑ |
| Increase $N_B$ | ↓ | ↓ | ↓ |
| Increase $\tau_E$ | ↑ | N.C. | ↑ |
| Increase $N_E$ | ↑ | N.C. | ↑ |